

# Midterm Project

314653005 嚴和育

October 2025

## 1 Introduction

In many scientific and engineering applications, one often needs to approximate a complex-valued function  $f(z)$  using a simpler model that can be efficiently and stably evaluated. Traditional polynomial interpolation methods, such as Taylor expansions, can suffer from numerical instability and large oscillations, especially for high degrees or nonsmooth functions.

The **adaptive Antoulas–Anderson (AAA) algorithm** addresses this by constructing rational approximations rather than polynomial ones. Rational functions represent functions with poles, asymptotes, or rapid variations with more accuracy using fewer terms. This algorithm provides an adaptive way to build such rational approximants directly from sampled data.

## 2 Problem Statement

For use the algorithm, we consider the problem for analytic functions in the unit disk.

Let  $\Delta = \{z \in \mathbb{C} : |z| \leq 1\}$ , and  $f$  be analytic on  $\Delta$ . Our goal is to see that  $f$  can be approximated in  $\Delta$ , and observe that  $f$  has a maximum error on different  $n$ , where  $n$  is degree; then we will consider whether  $f$  has the pole in some degree or not.

### 3 Method Explanation

First, the AAA algorithm constructs a rational function of the form

$$r(z) = \frac{\sum_{j=1}^m w_j f(z_j)/(z - z_j)}{\sum_{j=1}^m w_j/(z - z_j)}$$

where  $z_j$  are selected *support points*,  $f(z_j)$  are the corresponding function values, and  $w_j$  are *barycentric weights* computed adaptively.

Then, starting from a given sample set  $Z \subset \mathbb{C}$  and corresponding to data  $f(Z)$ , the process for iteration is :

- Choose a new support point  $z_m$  where the approximation  $r(z)$  has the largest error.
- Computes new weights  $w_j$  by solving a least-squares problem involving the matrix:

$$A_{ij} = \frac{f(z_i) - f_j}{z_i - z_j}.$$

- Updates the rational approximation and repeats until the residual falls below a suitable tolerance and test the convergence criterion, and its complexity  $O(M)$

In the process, it will use many linear algebra's tools, such as SVD and matrix computation, so it is hard for to do only by hand-writing when  $n$  approaches to infinity.

### 4 Simple Experiment

In the paper, the example test  $f(x) = \tan(x)$ , the poles happen at  $x = \pm(\pi/2)$ , if we set  $f(x)$  at  $M=128$  on the  $\Delta$ . Compute the degree and get the result from Python :

- A polynomial interpolant of degree  $n$
- A rational AAA approximant of type  $(n, n)$

Listing 1: AAA algorithm in unit disk

```
import numpy as np
import matplotlib.pyplot as plt
```

```

# Analytic function in the unit disk

M = 128
Z = np.exp(1j * 2 * np.pi * np.arange(M) / M) # 128 points on the
unit circle
F = lambda z: np.tan(z) # target function

# Compute AAA approximation (requires aaa() function)
r, z, f, w, err = aaa(F, Z)

# Evaluate error on dense test grid
Ztest = np.exp(1j * np.linspace(0, 2*np.pi, 1000))
error = np.abs(F(Ztest) - r(Ztest))

plt.figure(figsize=(12,5))

# --- Left subplot: function vs approximation
plt.subplot(1, 2, 1)
plt.plot(np.real(Ztest), np.real(F(Ztest)), 'b', label='Re[f(z)] =
Re[tan(z)]')
plt.plot(np.real(Ztest), np.real(r(Ztest)), '--r', label='Re[AAA
approx]')
plt.title('AAA Rational Approximation on Unit Disk')
plt.xlabel('Re(z)'); plt.ylabel('Re(f(z))')
plt.legend()
plt.grid(True)

# --- Right subplot: approximation error
plt.subplot(1, 2, 2)
plt.semilogy(np.arange(len(Ztest)), error, 'k')
plt.title('Approximation Error on the Unit Circle')
plt.xlabel('Sample Index')
plt.ylabel('|f(z) - r(z)|')
plt.grid(True)

plt.tight_layout()
plt.show()

```

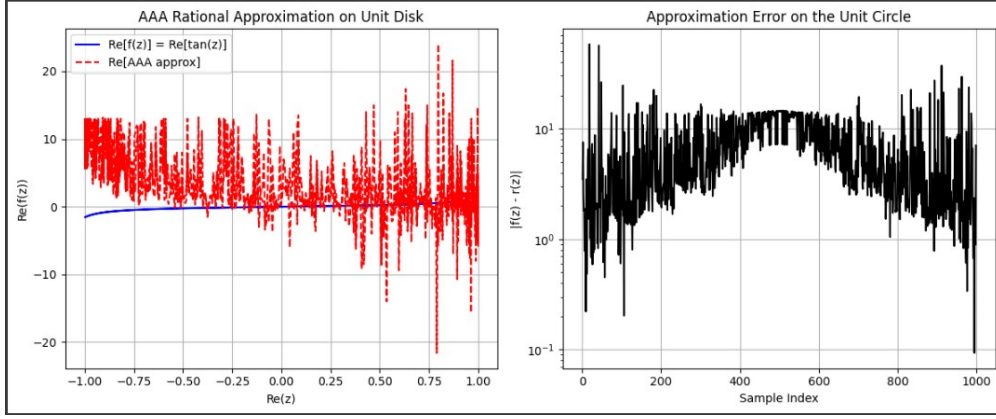


Figure 1: Function vs Approximation with Approximation Error

From above, it shows us that Rational approximations may be more efficient, we see that this example confirms numerical stability, and the figure shows the error plot decays rapidly.

- Number of support points used: about 100 ( the result is different from paper: 50)
- Final max : it runs to NaN, but i guess it is the impressively small error

However, it is not for sure that all of AAA approximation is no pole in  $\Delta$ , so we can adjust the degree of type  $(n, n)$  or setting  $n$  goes to infinity to conclude what condition about the singularity appears.

## 5 Summary and Conclusion

I think that we can test other algorithm for the same problem, or more conditons such as working on intervals, disconnected regions. Though this experiment shows that rational functions may have advantages over polynomials on the unit disk, but we still do not know this algorithm is suitable for every moment. Hence, it has several limitations to figure out :

- Preservation: Modified versions that maintain even/odd or real symmetry
- Consumption: Does not minimize any particular norm ? (like  $L^2$ )
- Numerically stable: If it is not barycentric representation, is it still stable ?
- Type  $(n, n)$  approximations: Extending to cases where numerator and denominator degrees differ.
- Matrix extensions: How about extends to matrix-valued functions ?

I think the AAA algorithm represents a significant advancement in practical rational approximation, and need to prepare more knowledge for analyzing and proof. And also, getting accurate rational approximation in any domains in computational mathematics is one of our dreams, maybe there have more efficient algorithms in the future.

## References

Nakatsukasa, Y., Sète, O., & Trefethen, L. N. (2018). The AAA algorithm for rational approximation. *SIAM Journal on Scientific Computing*, 40(3), A1494–A1522. <https://doi.org/10.1137/16M1106122>